

Experiment: Removal of Colour and COD from Wastewater using Activated Carbon

Reference standards: APHA 5220 (COD) · APHA 2120 (Colour) · ASTM D3860 (carbon adsorption evaluation)

1. Aim

To remove colour and chemical oxygen demand (COD) from a wastewater using powdered activated carbon; to determine the effect of carbon dose on removal efficiency; and, from the pooled class data, to establish the optimum dose.

2. Learning Outcomes

- Explain adsorption and distinguish it from absorption.
- Describe why activated carbon is an effective adsorbent.
- Measure colour and COD before and after treatment and calculate removal efficiency.
- Plot removal against dose and identify the optimum dose.
- Relate the laboratory result to the design of a tertiary carbon treatment stage.

3. Introduction & Background Concepts

Biological (secondary) treatment removes the biodegradable organic matter from wastewater, but it leaves behind two things that ordinary microbes cannot destroy: COLOUR, from dyes, syntans and natural tannins, and REFRACTORY COD — the fraction of the organic load that is not biodegradable. Both must be removed in tertiary treatment before discharge or reuse. Adsorption onto activated carbon is the classical and most widely used method of doing so.

What is adsorption?

Adsorption is the adhesion of dissolved molecules onto the SURFACE of a solid. The dissolved substance being removed is called the adsorbate; the solid it sticks to is the adsorbent. It is important to distinguish two similar words:

- **Adsorption** — molecules stick to the SURFACE of the solid (like dust settling on a wall).
- **Absorption** — molecules penetrate INTO the bulk of the material (like water soaking into a sponge).

In carbon treatment, the pollutant is adsorbed — held on the vast internal surface of the carbon particles. Nothing is chemically destroyed; the pollution is TRANSFERRED from the water onto the solid, which must then be regenerated or disposed of.

Why activated carbon works

Activated carbon is made by charring a carbon-rich material (coconut shell, wood, coal, or even leather waste) and then ‘activating’ it with steam or chemicals at high temperature. Activation burns out a vast network of internal pores, giving an enormous internal surface area:

- **Surface area** — typically 500–1,500 m² per gram. A single teaspoon of activated carbon can have the surface area of a football field.
- **Pore structure** — macropores act as highways carrying molecules inward; micropores (< 2 nm) provide the actual adsorption sites.

- **Surface chemistry** — the carbon surface is largely non-polar, so it strongly attracts the large, non-polar organic molecules that make up dyes and refractory COD.

This is precisely why carbon is so effective on colour: dye and tannin molecules are large, aromatic and poorly soluble — exactly the molecules that adsorb most strongly.

Forces involved

- **Physical adsorption (physisorption)** — weak van der Waals attraction; reversible, so the carbon can be regenerated. This dominates in water treatment.
- **Chemical adsorption (chemisorption)** — actual chemical bonding to the surface; stronger and largely irreversible.

PAC and GAC

Activated carbon is supplied in two forms:

Form	Powdered (PAC)	Granular (GAC)
Particle size	< 0.1 mm (fine powder)	0.6 – 4 mm granules
How it is used	Dosed into the water, mixed, then filtered off	Packed in a fixed-bed column; water flows through
Contact	Batch — fast, high surface exposure	Continuous flow
Regeneration	Usually not regenerated	Thermally regenerated and reused
Used in this experiment	YES — batch dosing is ideal for teaching	Industrial full-scale practice

Breakthrough and exhaustion

Adsorption capacity is finite. As carbon takes up more pollutant its remaining capacity falls, until it is saturated (exhausted) and removes nothing further. In a GAC column the point at which pollutant begins to appear in the outlet is called BREAKTHROUGH, and the carbon must then be regenerated or replaced.

4. Principle of the Experiment

A fixed volume of wastewater is contacted with a measured dose of powdered activated carbon and mixed for a fixed contact time. Colour-causing and organic molecules diffuse to the carbon surface and are adsorbed. The carbon is then separated by filtration, and the treated water is analysed for colour and COD. Comparing treated with untreated water gives the removal efficiency for that dose. Because each group uses a DIFFERENT dose, the pooled class results show how removal varies with dose, allowing the optimum dose.

5. Definitions & Key Terms

Adsorption — adhesion of dissolved molecules onto the surface of a solid.

Adsorbent — the solid that does the adsorbing — here, activated carbon.

Adsorbate — the substance being removed — here, colour bodies and organic (COD) molecules.

Activated carbon — a highly porous carbon with a very large internal surface area (500–1,500 m²/g), produced by carbonising and activating a carbonaceous raw material.

COD (Chemical Oxygen Demand) — the oxygen required to chemically oxidise all the organic matter in a sample, expressed in mg/L; it measures the TOTAL organic load, biodegradable and refractory alike.

Refractory COD — the fraction of COD that biological treatment cannot remove; the target of carbon adsorption.

Colour (true colour) — colour measured after filtering out suspended matter, reported in Hazen or Pt-Co units (or, here, as absorbance).

Carbon dose — mass of activated carbon added per litre of wastewater (g/L).

Contact time — the period the carbon is mixed with the water before separation.

Adsorption capacity (q_e) — mass of pollutant adsorbed per unit mass of carbon at equilibrium (mg/g).

Optimum dose — the lowest dose that achieves the required removal; beyond it, extra carbon gives little further benefit for the cost.

6. Group Allocation — Four Groups, Four Doses

All groups use the SAME wastewater and the SAME procedure, contact time and mixing speed. Only the CARBON DOSE differs. Each group also measures the untreated sample (0 g/L) as its control reference. Pooling the four results produces the dose–response curve and the isotherm.

Group	Carbon dose	Carbon in 500 mL	Members
Group 1	0.5 g/L	0.25 g	_____
Group 2	1.0 g/L	0.50 g	_____
Group 3	2.0 g/L	1.00 g	_____
Group 4	4.0 g/L	2.00 g	_____

Expected trend: removal rises steeply from Group 1 to Group 3, then levels off by Group 4 — the classic diminishing-returns shape that identifies the optimum dose.

7. Apparatus & Instruments

- **UV-Vis spectrophotometer** — to measure colour as absorbance
- **Photometric COD meter with digestion block** — to measure COD
- **Multi-parameter meter** — to measure pH and temperature
- **Magnetic stirrer with stir bars** — to provide the contact/mixing stage
- **Analytical balance (± 0.001 g)** — to weigh the carbon dose accurately
- **Beakers (500 mL), measuring cylinders, pipettes** — sample handling

- **Filtration set: funnel + Whatman No. 42 filter paper (or 0.45 µm syringe filters)** — to separate the carbon after contact
- **Stopwatch, spatula, wash bottle, cuvettes** — general

8. Materials & Reagents

- **Powdered activated carbon (PAC), laboratory grade** — the adsorbent
- **Wastewater sample** — biologically treated (secondary) effluent, or a coloured tannery/dye effluent
- **COD digestion vials (potassium dichromate reagent)** — for the COD test
- **Distilled water** — blanks, dilutions and rinsing
- **Dilute H₂SO₄ and NaOH** — for pH adjustment if required

Note on the carbon: dry the PAC at 105 °C for 1 hour and cool in a desiccator before use. Moist carbon gives a false (high) weight and therefore a wrong dose.

9. Sample Preparation

1. Collect a well-mixed sample of the wastewater; if it carries suspended solids, filter or settle it first — carbon adsorption is a POLISHING step and suspended matter would blind the carbon and confuse the colour reading.
2. Record the pH and temperature on the multi-parameter meter.
3. If the pH is strongly alkaline (> 9), adjust to approximately pH 6–7 with dilute acid and note the adjustment. Adsorption of organic molecules is generally better at slightly acidic to neutral pH.
4. Divide the prepared sample so that every group works from the SAME batch.

10. Detailed Procedure

Part A — Characterise the untreated wastewater (all groups, jointly)

1. Colour — find the wavelength: on the UV-Vis, scan a filtered portion of the raw sample across 400–700 nm and identify the wavelength of maximum absorbance (λ_{max}). Record it. All colour readings, before and after, must use this same wavelength.
2. Zero the spectrophotometer with a distilled-water blank, then measure and record the initial absorbance, A_0 , of the raw sample. If the reading exceeds the linear range (about 1.0 absorbance units), dilute the sample and record the dilution factor — then use the same dilution for every later reading.
3. COD — measure the initial COD, C_0 , of the raw sample using the photometric COD method (digest at 150 °C for 2 hours, cool, then read). Record it in mg/L.
4. Record initial pH and temperature.

Part B — Carbon contact (each group, its own dose)

1. Measure exactly 500 mL of the prepared wastewater into a clean 500 mL beaker and add a magnetic stir bar.
2. Weigh your group's dose of dried PAC accurately on the analytical balance (**0.25 g, 0.50 g, 1.00 g or 2.00 g** for 0.5, 1.0, 2.0 or 4.0 g/L respectively in 500 mL).

3. Add the carbon to the beaker all at once and start the stopwatch immediately.
4. Mix on the magnetic stirrer at a moderate, constant speed (about 150 rpm — fast enough to keep all the carbon suspended, not so fast as to splash) for a contact time of **60 minutes**. Keep the speed and time identical to the other groups.
5. At the end of the contact time, stop the stirrer and let the carbon settle for 5 minutes.

Part C — Separation and analysis

1. Filter the entire contents through Whatman No. 42 paper (or a 0.45 μm syringe filter) into a clean beaker. *Complete removal of carbon fines is essential — any carbon left in suspension will scatter light and give a falsely HIGH absorbance.*
2. Discard the first 10–15 mL of filtrate (it may carry fines), then collect the rest for analysis.
3. Measure the treated absorbance, A, at the same λ_{max} and same dilution as in Part A.
4. Measure the treated COD, C, by the same photometric method as in Part A.
5. Record the final pH.
6. Run a duplicate if time permits, and report the average.

11. Observations

A. Common data (untreated wastewater)

Parameter	Value
Wavelength used, λ_{max} (nm)	
Initial absorbance, A_0	
Initial COD, C_0 (mg/L)	
Initial pH / temperature	

B. Your group's data

Parameter	Value
Group number / assigned dose (g/L)	
Mass of carbon weighed (g)	
Sample volume (mL)	
Contact time (min) / stirring speed (rpm)	
Treated absorbance, A	

Parameter	Value
Treated COD, C (mg/L)	
Final pH	
Colour removal (%)	
COD removal (%)	

D. Pooled class results (copy every group's result)

Dose (g/L)	Absorbance A	Colour removal %	COD (mg/L)	C	COD removal %	q _e (mg/g)
0 (raw)		—			—	—
0.5 (Gr 1)						
1.0 (Gr 2)						
2.0 (Gr 3)						
4.0 (Gr 4)						

12. Calculations

A. Removal efficiency

$$\text{Colour removal (\%)} = [(A_0 - A) / A_0] \times 100$$

$$\text{COD removal (\%)} = [(C_0 - C) / C_0] \times 100$$

where A₀ and C₀ are the initial absorbance and COD, and A and C the values after treatment.

B. Adsorption capacity

The mass of pollutant taken up by each gram of carbon at equilibrium:

$$q_e = (C_0 - C_e) \times V / m \quad (\text{mg/g})$$

where C₀ = initial COD (mg/L), C_e = equilibrium (treated) COD (mg/L), V = volume of sample in litres, and m = mass of carbon in grams.

13. Graphs Required

- Graph 1 — Removal (%) versus carbon dose (g/L): plot both colour and COD removal on the same axes. The optimum dose is where the curve flattens.

14. Precautions

- Dry the activated carbon before weighing; moist carbon gives an incorrect dose.

- Keep contact time, stirring speed and sample volume identical across all four groups — only the dose may differ.
- Filter completely before measuring absorbance; suspended carbon fines scatter light and inflate the reading.
- Use the same wavelength and the same dilution factor for every colour measurement.
- Discard the first portion of filtrate, which may carry fines.
- Handle COD digestion vials with care: they contain strong acid and dichromate, and are digested hot.
- Rinse cuvettes with the sample before filling, and wipe the outside with lint-free tissue.
- Record the final pH — adsorption performance is pH-dependent.

15. Safety

- Wear a lab coat, gloves and safety goggles throughout.
- COD reagent contains concentrated sulphuric acid, potassium dichromate (a carcinogen) and mercury or silver salts — do not open hot vials; allow them to cool fully.
- Dispose of used COD vials as hazardous chemical waste; never pour them down the drain.
- Powdered activated carbon is a fine dust — weigh it carefully to avoid inhaling it, and clean spills with a damp cloth rather than by brushing.
- Treat all wastewater as potentially contaminated; wash hands after handling.

16. Results

Report the following: the initial colour (absorbance) and COD; your group's treated values and removal percentages; the pooled dose–response table; the optimum dose is identified.